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Dance of two brains: Interval subdivision in alternated condition enhances resistance to interference by others

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ABSTRACT

The accomplishment of interpersonal sensorimotor synchronization is a challenging endeavor because it requires the achievement of a balance between accurate temporal control within individuals and smooth communication between them. This raises a critical question: How does the brain comprehend and process the perceptual information of others to guarantee accurate temporal control of action goals in a social context? A joint synchronization - continuation tapping task was conducted together with varying relative phases (0°/180°) and intervals of tempos (400 ms/800 ms/1600 ms) while neural data was collected using fNIRS (functional near-infrared spectroscopy). Individuals showed better behavioral performance and greater interpersonal brain synchronization(IBS) in the left dorsolateral prefrontal cortex at alternated condition (180° relative phase) compared to symmetric condition (0° relative phase), suggesting that the individual can better maintain behavioral performance and show improved IBS when the partner taps between the individual's gaps. Meanwhile, in most levels of alternated condition, IBS is inversely proportional to interference from partner, implying the counteraction of IBS against interference from others. In addition, when the interval of tempo was 1600 ms, behavioral performance showed a sharp decline, accompanied by a decrease in IBS, reflecting that IBS in SMS reflects effective information exchange between individuals rather than ineffective interference with each other. This study provides insight into the mechanisms underlying sensorimotor synchronization between individuals.

1. Introduction

Sensorimotor synchronization (SMS) with others is widely present in daily life. In order to achieve shared goals, whether consciously or unconsciously, individuals need to adjust the timing of their actions and cooperate in collective behavior (Schmidt et al., 1990). For example, audiences in concert halls tend to clap in unison. When two people sit side by side on chairs and swing their legs, they tend to synchronize. In a marching band, each individual can play while moving forward in the same rhythmical pace. In the context of complex social situations, the attainment of team goals requires collaboration between individuals and other team members. Each member in this process needs to receive information from other team members to adjust or guide their actions for coordination. However, external information may be disruptive and does not always serve as a reference. This questioning is often subconscious but enough to influence their planning and decision-making, resulting in decreased task performance. It may also lead individuals to execute actions earlier in the next attempt to achieve simultaneous execution with their partners (Repp, 2005). However, reference information and disruptive information are often difficult to separate. As in an orchestra, musicians face challenges in maintaining their pace while synchronizing with other performers due to the tempo of the music (Kimura et al., 2020), which shows that even skilled performers cannot wholly avoid interference from others (Wolf and Knoblich, 2022). The reason is that we cannot share the same brain, which means that individuals need to comprehend the intentions of others, predict their actions, and adjust their own movements accordingly to achieve shared goals. However, there is still a lack of understanding of how our brain perceives and balances internal timing with sensorimotor information generated by others to achieve interpersonal SMS.

The interpersonal SMS requires cooperation and mutual attention between everyone, which makes all participants act as a group to accomplish a common goal, this synergy is defined as entrainment (Davis, 1982; Newtson et al., 1987), which is defined as spatiotemporal

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coordination resulting from perceived spatial and timing signals (Phillips-Silver et al., 2010). Individuals may be influenced separately or together by multiple variables related to others, such as relative phase, tempo, sensory channels, etc. Each member involved in entrainment interacts with these kinds of sensorimotor information with attention and intention, giving perceptual signals social properties for mutual understanding and effective entrainment (Cross et al., 2019; He et al., 2016; Rosso et al., 2021; Rosso et al., 2023). To achieve entrainment, there are two hierarchical coordination modes, which are emergent and planned coordination. They can work together for joint actions, which are important for maintaining one's own actions and coordinating with others(Knoblich et al., 2011). However, keep balance in social context is complex and hard to understand separate mode. The impact of the interaction between these two modes on individual tendencies to focus on themselves or collective action remains complex and unclear.

Individuals in the entrainment are correlated in time series, specifically, maintain regular relative phases. 0° (symmetric condition) and 180° (alternated condition) are two common and easily maintained relative phases. For example, when we walk side by side with others (symmetric mode), and during a partner waltz, one person steps forward while the other steps back (alternated condition). In contrast, symmetric condition (0° relative phase) is easier to maintain compared to alternated condition (180° relative phase). Besides, alternated condition may become challenging to maintain and eventually collapse into a 0° phase as the tempo speeds up (Bressler and Kelso, 2001; Haken et al., 1985; Kelso, 2021; Schoner and Kelso, 1988; Zanone and Kelso, 1992).

However, studies in the field of temporal perception found that individuals maintain their actions more stable when intervals are subdivided, which is similar to the role of reference and guidance (Ono et al., 2022; Repp, 2003, 2008; Repp and Jendoubi, 2009). When playing in an ensemble, it is challenging to synchronize with others immediately after a long note, particularly when there are no cues from an outside conductor. Subdividing into smaller units for long intervals may facilitate this process (Repp, 2010). The mechanism by which interval subdivision occurs is to reduce the perceived inter-stimulus onset intervals (IOI) because the variance of the inter-tap interval (ITI) increases nonlinearly as the IOI grows. This results in the sum of the variances of the two intervals generated by subdivision is smaller than the variance of the sum of the two intervals (Getty, 1975, 1976), which results in a benefit. And the setup of taking turns tapping between individuals in alternated conditions may objectively lead to interval subdivision. This differs from the explanation of relative phase in the previous views. This study could be practical significance in clarifying these confusions and improving and promoting the theory of dynamics coordination (Haken et al., 1985; Kelso, 2021; Schmidt et al., 1990, 2011), which is related to the maintenance and variability of the rate and relative phase of the periodic movement.

Entrainment is not exclusively determined by relative phase, but tempo can also have an impact. As the tempo slows down or becomes more challenging, individuals rely less on self-correction of their perceived movement synchronization within a wider range. This can increase the disruptive effects of tempo by others (Kimura et al., 2020; Okano et al., 2017b), named model of joint rushing (Konvalinka, 2009; Okano et al., 2017a). For example, people engage in joint rhythmic activity unintentionally quicken their tempo. The greater the variability of asynchrony, the stronger the joint rushing tends to be (Konvalinka et al., 2009; Okano et al., 2017b; Wolf et al., 2019), which can weaken the performance of SMS between individuals. And joint rushing can also be observed in more natural social interactions (Wolf et al., 2023). However, if interval subdivision can be effective for the reduction of variability in dyadic condition, the relative phase may interact with tempo as joint rushing emerges. That is, in the alternated condition, individuals could at least partially counteract the effects of increased variability from joint rushing by interval subdivision via others' tapping, whereas in the symmetric condition this counteracting effect could not be observed because of the absence of interval subdivision. Furthermore, it is unclear whether this offsetting of disruption in the alternated condition eliminates the difference in disruption compared to the individual condition, or may even lead to superior stability compared to the separate condition. In that case, choosing a 180° relative phase as the preferred phase for joint actions becomes necessary.

Behavioral entrainment is controlled by neural entrainment at the level of the brain, which involves unidirectional synchronization of neural oscillations to an external rhythmic stimulus (Rosso et al., 2023). In addition to observing the mutual influence in SMS from a behavioral standpoint, exploring the underlying mechanisms of the interactive process from a neural perspective can provide a more direct and objective view. Interpersonal Brain Synchronization (IBS) has been used in recent years as an indicator to evaluate the effectiveness of interpersonal interactions. Higher levels of IBS can reflect better teacher-student interactions (Liu et al., 2019; Zheng et al., 2018a), more synchronized responses (Cui et al., 2012; Pan et al., 2017), and more efficient problem-solving (Lu and Hao, 2019), among other factors. It is important to note that in the context of interpersonal SMS, simply engaging in the same activities does not necessarily induce IBS, which requires joint attention and shared goals (Dai et al., 2023; Hu et al., 2022; Kikuchi et al., 2021; Wang et al., 2020b; Xie et al., 2021). According to the mutual prediction theory, social interaction motivates and enables the integration of the brain and behavior into a unified model in which the social brain is working to predict and enact social behavior (Hamilton, 2021). The presence of interval subdivision effects between individuals provides the possibility for such mutual prediction; if not, interpersonal brain synchronization will not develop. This feature can help distinguish whether an individual's performance decreases due to the presence of partners in the current process, or if the presence of a common goal allows individuals to maintain or improve their performance in SMS with others. Therefore, it is necessary to incorporate cross-brain measures in studying interpersonal SMS. Through further research and understanding of the mechanisms underlying interpersonal information exchange, we can better apply it to promote social interaction and promptly identify the impairment of individual behavioral performance caused by SMS.

Given the challenge of simultaneously investigating both behavioral performance and the allocation and exchange of cognitive resources in the brain, the involvement of neuroimaging techniques may help to better understand the underlying mechanisms of temporal processing in social environments. In terms of brain region selection, to observe the synchronization and interference of tempo information between individuals, it is essential to focus on the regions responsible for timing and movement and the regions involved in social interaction and action observation. The pre-supplementary motor area (pre-SMA) serves as the timing component of the simulated action (Cannon and Patel, 2021). Research has shown that the pre-SMA also plays a crucial role in visuomotor sequence learning(Shimizu et al., 2020) and cognitive flexibility (Mukherjee et al., 2021). It is also part of a mirror neuron network involving the inferior parietal lobule (IPL) (Cattaneo and Rizzolatti, 2009; Moretti, 2016) and dorsolateral prefrontal cortex (dlPFC) is responsible for top-down attentional control and attentional flexibility (Tei et al., 2017; White et al., 2023; Wu et al., 2014). Research has shown that the dlPFC plays a crucial role in directing visual attention to important information while avoiding distractions(Carlisle, 2019; Liu et al., 2023; Stenson et al., 2023). Research by Watanabe et al. (2018) found a positive correlation between dlPFC activation and self-reported cognitive effort, suggesting that dlPFC activation is related to perceived cognitive demand. Specifically, these ROIs correspond to functions identified by previous research, such as cognitive flexibility, the mirror neuron system, and the potential use of working memory, which may be recruited in the tasks set in this study. Therefore, we divided and selected channels based on prior literature to guide our study.

2. Method

This study employed the synchronization-continuation (SC) tapping task, in which participants tapping synchrony with a presented beat, as the experimental paradigm to investigate tempo maintenance performance in a dyadic setting. This involved two variables: tempo and relative phase. Although previous studies have indicated that auditory SMS is more effective than visual SMS. For example, parkinsonian gait can be better improved when cued by an external auditory tempo rather than a visual stimulus (Arias and Cudeiro, 2008; Rochester et al., 2005). Auditory timing compared to visual timing tasks often yield more activation in pre-SMA and premotor cortex (Jancke et al., 2000). However, considering that the focus of this study is not on the differences in sensory channels, meanwhile studies by Cui et al. (2012) demonstrated that visual stimuli can induce IBS, which is helpful for our study on interpersonal SMS. Therefore, we still used visual stimuli such as blinking dots as rhythmic presentations.

2.1. Participants

According to G*power, the minimum sample size required is 17.8 (effect size $f = 0.25$, α err prob $= 0.05$, power $= 0.8$). This study recruited 36 participants (18 pairs) with an average age of 22.61 ± 2.40 years, who were recruited through online social media platforms or offline poster advertisements. There were 11 male and 25 female participants, including 11 pairs of the same gender and 7 pairs of different genders. The two participants in each pair were not acquainted with each other. All participants were right-handed, had normal or corrected-to-normal vision, and had no history of mental disorders. Prior to the experiment, informed consent was obtained from each participant. After completing the experiment, each participant received a payment of 50 RMB.

2.2. Task and procedure

This study employed a modified SC tapping task. Participants were seated facing separate monitors (Dell E2318H, 1920 × 1080, 60 Hz, 23 inch), with the monitors placed back-to-back. Before the start of all experiments, participants were asked to wear earplugs to reduce sound interference as much as possible. And the neural activity of the individual in a resting state is first collected using functional near-infrared spectroscopy (fNIRS), lasting for 3 minutes.

The experimental program was written using Psychopy software (version: 2021.2.3). At the beginning of each block, a fixation point was presented for 2 s. Participants were then instructed to learn the tempo based on the standard tempo displayed on their respective screens (indicated by blue dot flashing, which is a 5 cm diameter circle). Note that in this stage, Participants were asked to start tapping once they have already learned the tempo. After the standard tempo had been presented ten times by flashing, it disappeared, and participants were required to continue tapping according to the learned tempo 30 additional times. There was a 15s break before moving on to the following tempo. Each kind of IOI in the single task contains four blocks (see Fig. 1).

For dyadic tasks, the standard tempo light and the feedback of their tapping were presented on different screens. For example, for participant A, the left side of the screen would present the standard tempo stimulus that he/she needs to learn (metronome stage) and feedback on the individual tapping (maintenance stage), while the right side of the screen would present the standard tempo stimulus that his or her partner needs to learn (metronome phase) and feedback on the partner's tapping (maintenance stage). For participant B the direction was reversed (after the experiment, all participants reported that they could see the blinking on the other half of the screen). After half of the experiment, the positions of the blue dots on the screens were exchanged for participants. All participants were instructed to maintain the standard tempo and synchronize with their partner according to the standard relative phase (Kimura et al., 2020; Okano et al., 2017b; Wolf and Knoblich, 2022). Three different IOI were used to represent different standard tempos, namely 400 ms, 800 ms, and 1600 ms. This choice was based on Miyake et al. (2004), who suggested that as the IOI increase, individuals gradually shift from automatic processing of time prediction to requiring active allocation of attentional resources for time prediction. Two kinds of relative phases existed on the screen: symmetric condition (0° relative phase) and alternated condition (180° relative phase). In the dyadic task, the different relative phases and IOIs together form six conditions, each with a total of four blocks. All participants needed to complete the task together with their partner and individually, and the order of single and dyadic tasks and the order of the two participants in the single task were balanced across all participants (see Fig. 2).

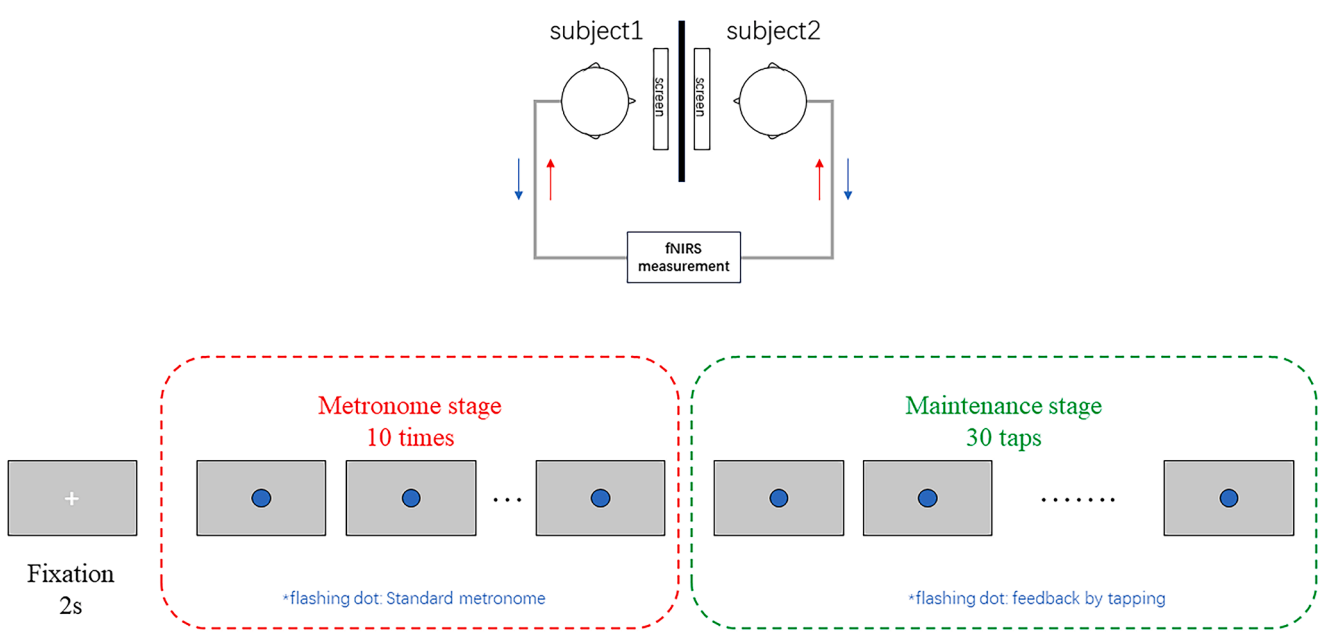

Fig. 1. Participants were seated facing separate monitors, with the monitors placed back-to-back (above); task procedure for a single task(below).

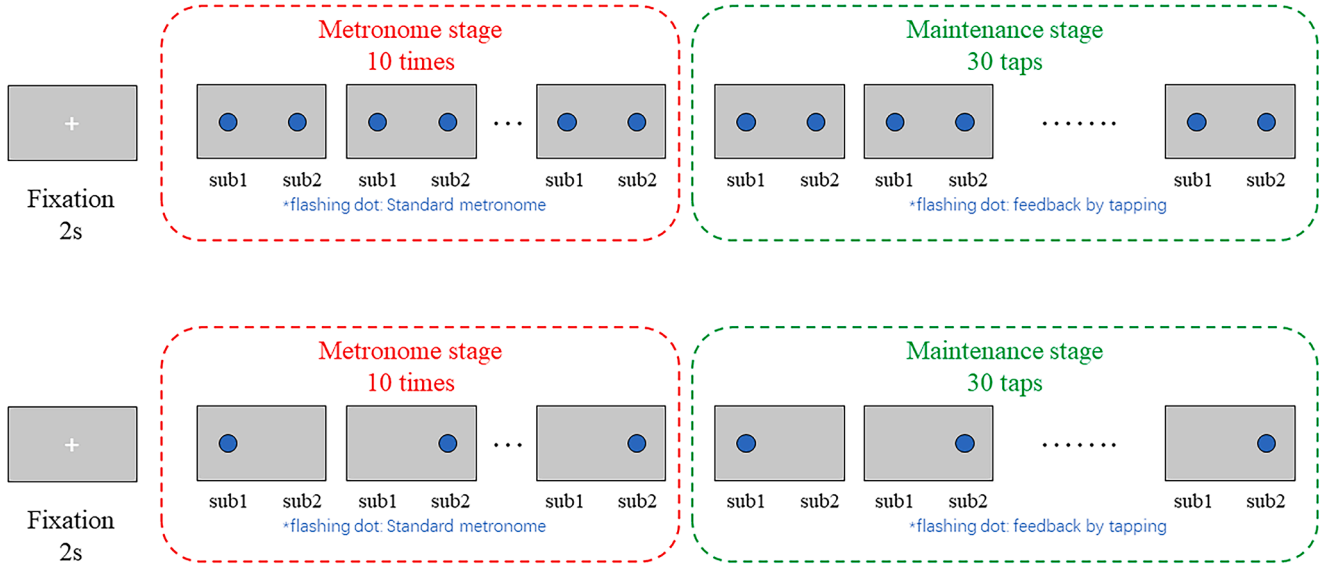

Fig. 2. Task procedure for dyadic symmetric (above) and alternated condition (below).

2.3. fNIRS data acquisition

We used a functional near-infrared spectroscopy (LABNIRS, including 780 nm, 805 nm, and 830 nm) from Shimadzu, Japan, changes in the oxygenated hemoglobin concentration ([HbO]), deoxygenated hemoglobin concentration ([HbR]), and total hemoglobin concentration ([HbT]) in the detection region were obtained according to the modified Beer-Lambert Law (Cope et al., 1988). The sampling rate in this study was 27.7 Hz. According to prior knowledge, the left and right dorsolateral prefrontal cortex, the anterior supplementary motor area, and the left and right inferior parietal lobe were selected as the measurement areas. According to the 10-05 international lead system, the self-designed probe distribution was used (Cutini et al., 2011; Jurcak et al., 2007). There were 21 probes in each subject's measurement area, forming 27 channels (see Fig. 3 and Table 1).

2.4. Data analysis

2.4.1. Behavior data

We obtained the ITI time series by subtracting the tap onset time from the adjacent tap onset time (Eq. (1)). Then, we performed outlier detection and rejection in terms of both intra-individual ITI and inter-individual ITI difference. For all conditions, if the absolute value of ITI exceeds 1/4 of IOI, that set of tap data will be excluded. Further, as for dyadic condition, if absolute value of ITI difference between participants exceeds the IOI of 1/4 of the required difference (symmetric condition: 0; alternated condition: 1/2 IOI). The 29 ITIs generated across all conditions and blocks for each subject group were tested according to the above criteria, with an average pass rate of 92.1 % (an average of 26.7 conforming ITIs in each block). Non-compliant ITIs was eliminated and interpolated by the average of the other ITIs in that block. Afterward, the difference between the individual's ITI and the corresponding IOI will be divided by the IOI for standardization (Eqs. (2) & (3)) as individual deviation. The further away from 0 this index is , the further the individual deviates from IOI. Individual deviations will be further employed for statistical analysis as response variable.

$$ITI(n) = t(n+1) - t(n) \tag{1}$$

$$\Delta ITI_{avg} = \frac{\sum_{n=1}^{29} ITI(n)}{29} \tag{2}$$

$$Individual\ Deviation = \frac{\Delta ITI_{avg} - IOI}{IOI} \tag{3}$$

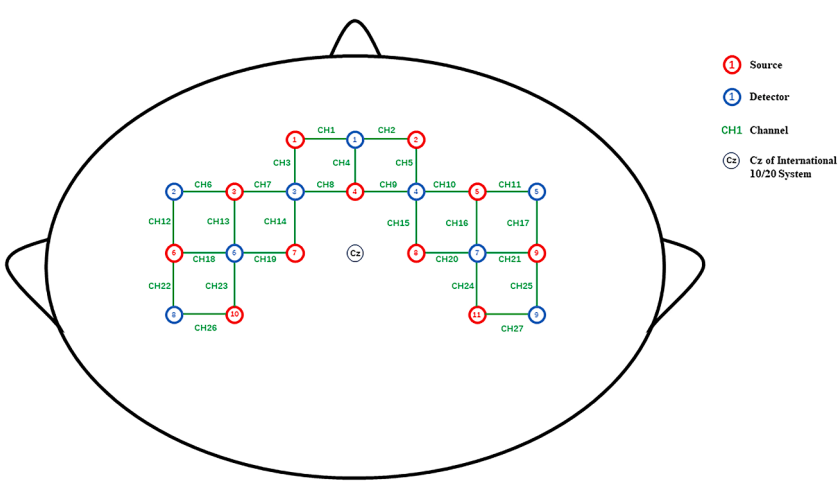

Fig. 3. Spatial profile of fNIRS probes. Red circles indicate optical sources. Blue circles indicate detectors. Green numbers (1–27) indicate fNIRS channels. The optical sources and detectors were on the international 10–05 standard positions.

Table 1
Locations on the international 10-05 system, as well as the corresponding anatomical labels, region of interest and MNI coordinates for the placement of fNIRS channels.

Channel	Position	ROI	Region	Lateralization	MNI coordinates		
					x	*y*	*z*
1	F1h	1	dlPFC	L	−10	45	51
2	F2h	2	dlPFC	R	13	45	51
3	FFC1	3	pre-SMA	-	−23	32	55
4	FFCz	3	pre-SMA	-	2	31	58
5	FFC2	3	pre-SMA	-	26	31	56
6	FC5h	1	dlPFC	L	−56	14	33
7	FC3h	1	dlPFC	L	−39	17	56
8	FC1h	3	pre-SMA	-	−13	18	67
9	FC2h	3	pre-SMA	-	15	17	67
10	FC4h	2	dlPFC	R	40	18	56
11	FC6h	2	dlPFC	R	58	13	33
12	FCC5	1	dlPFC	L	−65	−2	24
13	FCC3	3	pre-SMA	-	−52	0	50
14	FCC1	3	pre-SMA	-	−28	3	68
15	FCC2	3	pre-SMA	-	28	2	67
16	FCC4	3	pre-SMA	-	54	0	50
17	FCC6	2	dlPFC	R	66	−3	24
18	C5h	4	IPL	L	−63	−17	41
19	C3h	4	IPL	L	−43	−16	65
20	C4h	5	IPL	R	44	−15	64
21	C6h	5	IPL	R	65	−17	41
22	CCP5	4	IPL	L	−67	−31	29
23	CCP3	4	IPL	L	−53	−30	54
24	CCP4	5	IPL	R	57	−30	55
25	CCP6	5	IPL	R	69	−30	29
26	CP5h	4	IPL	L	−61	−43	44
27	CP6h	5	IPL	R	62	−43	45

Furthermore, we aim to observe the extent to which individuals influence each other. Firstly, we obtained the ITI time series by subtracting the tap onset time from the adjacent tap onset time (Eq. (1)). Then, average all intervals across all blocks for each condition to obtain the average ITI for each condition. By dividing the difference between the average ITI in the paired condition (symmetric / alternated) and the solo condition by average ITI in the solo condition, we can derive a new variability caused by the presence of others, defined as the interference from partner (Eq. (4)), which was employed as response variable for statistical analysis. All ANOVAs were performed using Jamovi 2.4.1 (R Core Team, 2022; The jamovi project, 2023).

$$Interference\ From\ Partner = \frac{ITI_{avg(paired)} - ITI_{avg(solo)}}{ITI_{avg(solo)}} \quad (4)$$

2.4.2. fNIRS data analysis

First, we conduct an initial assessment of the signal quality of the channels and exclude channels with a coefficient of variation greater than 15 % (Niu et al., 2019; Piper et al., 2014). Then, the raw optical intensity signals are converted into optical density signals. Subsequently, a principal component analysis (PCA) filtering technique is employed to remove the global physiological noise in the fNIRS signals (Huppert et al., 2009). Specifically, the process begins by performing PCA on the three original waves of different wavelengths, decomposing them into eigenvectors and eigenvalues. The variance percentage of each principal component is calculated based on the obtained eigenvalues. The minimum number of principal components is then determined such that the sum of the variance percentages exceeds 80 %. This number is chosen as the retained principal component count. Finally, the data is reconstructed using the retained principal component count to obtain the filtered result. Then, the optical density signals are transformed into hemoglobin oxygenation signals. Additionally, a correlation-based signal improvement method (CBSI) is utilized to eliminate head motion artifacts based on the principle that the concentration changes of oxygenated and deoxygenated hemoglobin should be negatively correlated (Cui et al., 2010), which involving subtracting the mean values of [HbO] and [HbR] to obtain their respective changes, calculating the standard deviations of the two indicators, determining the ratio factor between the two standard deviations, and finally adjusting HbO based on this ratio factor to achieve motion correction. Due to the proven sensitivity of HBO signals to changes in cerebral blood flow during fNIRS measurements, we focused on the HBO time series (Chen et al., 2018; Cui et al., 2012; Hoshi, 2007; Niu et al., 2013, 2019; Pan et al., 2017; Wang et al., 2021, 2020a)

All time series of HBO signal were resampled at 1 Hz to facilitate the subsequent calculations. We used wavelet transform coherence (WTC) to assess the relationship between each pair of subjects for the HbO time series data in all the blocks. In order to select the interest frequency band, we conducted paired-sample t-tests on the task state and resting state, averaging across all conditions and blocks on a frequency-by-frequency basis. We identified a frequency range (20-42s, or to say 0.024–0.050 Hz) in which the WTC values during the task condition were greater than those during the resting state. No further correction was applied because this analysis was solely used to identify patterns along the frequency range rather than to obtain final results (Lu et al., 2019b; Zheng et al., 2018b). This frequency range was within the overall range covered by all task conditions (12-48.8s, or to say 0.021 – 0.083 Hz, See supplementary materials 1 and 2).

We calculated wavelet coherence on the [HbO] of two individuals in all blocks of the dyadic task. Then, the results across both the time and frequency dimensions for each block were averaged, obtaining task-specific WTC values. By subtracting the WTC values during the resting state, we obtained IBS, which represents the synchronization between individuals. Afterward, we performed a Fisher-Z transformation on the IBS to meet the normal distribution requirement.

3. Result

3.1. Behavioral performance—Individual deviation

The individual deviation was used to conduct a 3 × 3 repeated-measures analysis of variance (ANOVA), based on a 3 (relative phase: single/symmetric/alternated) × 3 (IOI: 400 ms/800 ms/1600 ms) experimental design. The main effects of relative phase ($F_{2,70} = 12.561$, $p < 0.001$, $\eta_p^2 = 0.264$) and IOI ($F_{2,70} = 29.694$, $p < 0.001$, $\eta_p^2 = 0.459$) were significant. Furthermore, the two-way interaction was also significant ($F_{4,140} = 2.455$, $p = 0.049$, $\eta_p^2 = 0.066$). *Post hoc* analysis revealed that, for the relative phase, the individual deviation was significantly lower in single condition compared to both symmetric and alternated condition ($t_{35} = 4.921$, $p < 0.001$; $t_{35} = 2.394$, $p = 0.022$), and individual deviation in alternated condition was lower than symmetric condition ($t_{35} = 2.730$, $p = 0.010$) (see supplementary material 3). P-values for multiple comparisons had been adjusted using false discovery rate (FDR) correction.

Regarding IOI, individual deviation in 1600 ms was significantly higher compared to 400 ms ($t_{35} = 5.891$, $p < 0.001$) and 800 ms ($t_{35} = 5.891$, $p < 0.001$). However, there's no difference between 400 ms and 800 ms ($t_{35} = 2.338$, $p = 0.064$).

Simple effect analysis for two-way interaction revealed that in 400 ms, there was no difference between all three levels of relative phase (single-symmetric: $t_{35} = 1.968$, $p = 0.076$; single-alternated: $t_{35} = 2.120$, $p = 0.057$; symmetric-alternated: $t_{35} = -1.077$, $p = 0.336$). Nevertheless, in the 800 ms, there was also no difference between the single and alternated conditions, while individual deviation in these two conditions were all lower than that in the symmetric condition (single-symmetric: $t_{35} = 4.969$, $p < 0.001$; single-alternated: $t_{35} = 1.605$, $p = 0.151$; symmetric-alternated: $t_{35} = -3.586$, $p = 0.021$). Furthermore, in 1600 ms, the relationship between these three conditions was similar to that in 800 ms (single-symmetric: $t_{35} = 3.181$, $p = 0.006$; single-alternated: $t_{35} = 1.262$, $p = 0.259$; symmetric-alternated: $t_{35} = -2.842$, $p = 0.013$, see Fig. 4). P-values for multiple comparisons had been adjusted using FDR correction.

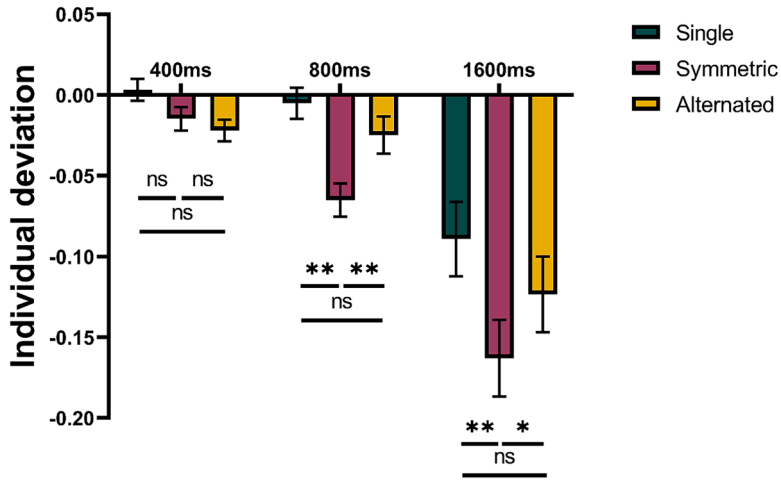

Fig. 4. In 400 ms, there was no difference between all three levels of relative phase. However, in the 800 ms condition, there was also no difference between the single and alternated conditions. In contrast, the individual deviation in these two conditions was all lower than that in the symmetric condition. Moreover, in 1600 ms, the relationship between these three conditions was similar to that in 800 ms. * $p < 0.05$. ** $p < 0.01$. *ns No significance.*

3.2. Behavioral performance—Interference from partner

The interference from partner was used to conduct a 2 × 3 repeated-measures ANOVA, based on a 2 (relative phase: symmetric/alternated) × 3 (IOI: 400 ms/800 ms/1600 ms) experimental design. The main effect of relative phase ($F_{1,17} = 6.150$, $p = 0.024$, $\eta_p^2 = 0.266$) was significant. *Post hoc* analysis revealed that interference from partner was significantly higher in symmetric condition compared to alternated condition ($t_{17} = 2.480$, $p = 0.024$) (see Fig. 5). However, the main effect of IOI ($F_{2,34} = 0.797$, $p = 0.459$, $\eta_p^2 = 0.045$) and two-way interaction was nonsignificant ($F_{2,34} = 2.828$, $p = 0.073$, $\eta_p^2 = 0.143$).

3.3. Interpersonal brain synchronization

First, we averaged all task-related conditions and their corresponding blocks under each channel. Then, according to previous research on the functional localization of the channels in this study, we divided them into five corresponding regions of interest (ROI) (left/right dlPFC, pre-SMA, and left/right IPL, see Table 1), and averaged all channels within each region of interest. This yielded the WTC values of the five ROIs for all participants. A one-way ANOVA with ROI as the independent variable was conducted to identify the region with the highest IBS for further statistical analysis. The result indicated a significant main effect of ROI ($F_{4,68} = 7.886$, $p < 0.001$, $\eta_p^2 = 0.317$). *Post hoc* analysis showed that the WTC value in the left dlPFC was higher than those in the other four brain regions. P-values for multiple comparisons had been adjusted using FDR correction. Therefore, it was chosen as the ROI for the subsequent statistical analysis (see supplementary material 4A and 4B).

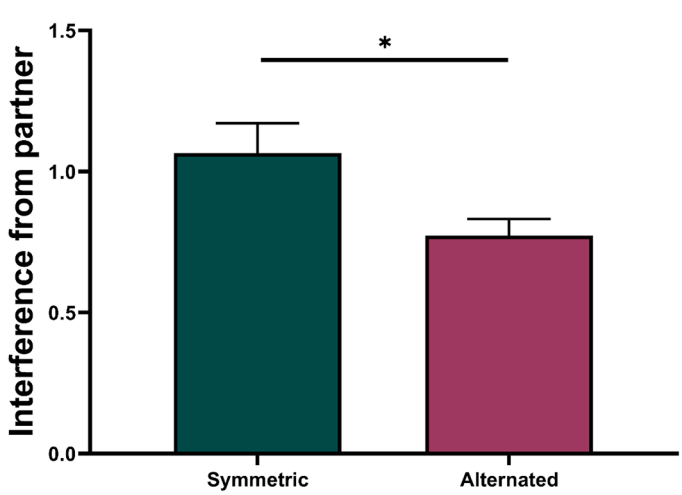

Fig. 5. Interference from partner was significantly higher in symmetric condition compared to alternated condition.

Then, a permutation test confirmed that IBS was specific to interactions occurring between real task partners, which means all real dyads were shuffled, resulting in 18 new pseudo-dyads, repeated 1000 times. Then one-sample t-test was conducted between IBS of all pseudo-dyads of all 1000 shuffled result and zero. The results showed that pseudo-dyads did not exhibit higher IBS during the task compared to rest periods ($t_{999} = 0.856$, $p = 0.392$), suggesting that the interactive nature of the joint SC task implemented in this study elicited IBS only in real dyads, reflecting the presence of interactions between them (see supplementary material 4C).

Afterward, we conducted a repeated measures ANOVA on the IBS of ROI−1, based on a 2 (relative phase: symmetric/alternated) × 3 (IOI: 400 ms/800 ms/1600 ms) experimental design. The results revealed a significant main effect of relative phase ($F_{1,17} = 4.547$, $p = 0.048$, $\eta_p^2 = 0.211$) and IOI ($F_{2,34} = 4.113$, $p = 0.025$, $\eta_p^2 = 0.195$). *Post hoc* analysis showed that the IBS at the alternated condition was higher for the relative phase than that at the symmetric condition ($t_{17} = 2.132$, $p = 0.048$) (see Fig. 6).

As for IOI, the IBS at 1600 ms was lower than those at 400 ms and 800 ms ($t_{17} = -2.870$, $p = 0.011$; $t_{17} = -2.264$, $p = 0.037$), while there was no significant difference between 400 ms and 800 ms ($t_{17} = 0.714$, $p = 0.485$). P-values for multiple comparisons had been adjusted using FDR correction. However, the two-way interaction effect was insignificant ($F_{2,34} = 0.366$, $p = 0.696$, $\eta_p^2 = 0.021$).

3.4. Cerebral activation

First, we compared the [HbO] of priori-measured brain regions to select regions of interest (ROI) that matched our task. Through this comparison, we found that [HbO] in the left dorsolateral prefrontal cortex (ROI-1) and left inferior parietal lobe (ROI-4) were higher than those in the other three brain regions, with no difference observed between the two regions of interest. P-values for multiple comparisons had been adjusted using FDR correction. Therefore, they were chosen as the ROI for the subsequent statistical analysis (see supplementary material 5). We conducted a repeated measures ANOVA on the [HbO] of ROI-1, based on a 3 (relative phase: single/symmetric/alternated) × 3 (IOI: 400 ms/800 ms/1600 ms) experimental design. The results revealed a

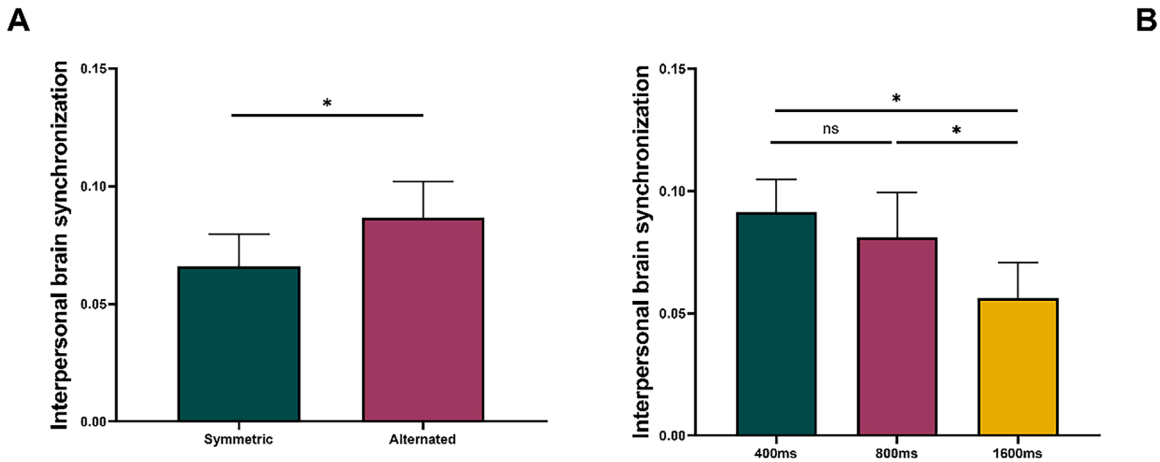

Fig. 6. **(A)** As for the relative phase, the IBS at the alternated condition was higher than that at the symmetric condition; **(B)** IBS at 1600 ms was lower than those at 400 ms and 800 ms, while there was no significant difference between 400 ms and 800 ms.

significant main effect of relative phase ($F_{2,70} = 18.379$, $p < 0.001$, $\eta_p^2 = 0.344$). *Post hoc* analysis showed that for the relative phase, [HbO] at the alternated condition was lower than that at the symmetric condition ($t_{35} = -2.208$, $p = 0.034$), and [HbO] at the single condition was lower than those at the symmetric ($t_{35} = -6.846$, $p < 0.001$) and alternated condition ($t_{35} = -3.403$, $p = 0.002$) (see Fig. 7). P-values for multiple comparisons had been adjusted using FDR correction. However, the main effect of IOI ($F_{2,70} = 0.074$, $p = 0.929$, $\eta_p^2 = 0.002$) and two-way interaction effect ($F_{2,70} = 1.441$, $p = 0.224$, $\eta_p^2 = 0.040$) were nonsignificant.

Then, A similar analysis was conducted for ROI-4. However, the main effects (relative phase: $F_{2,70} = 0.404$, $p = 0.669$, $\eta_p^2 = 0.011$; IOI: $F_{2,70} = 2.268$, $p = 0.111$, $\eta_p^2 = 0.061$) and two-way interaction effect ($F_{2,70} = 2.158$, $p = 0.077$, $\eta_p^2 = 0.058$) were both nonsignificant.

3.5. The IBS—Interference from partner relation

Pearson correlation analyses were conducted between significant IBS at ROI-1 and the interference from partner. Significant negative correlations were found at 400 ms and 800 ms under the alternated condition (alternated 400 ms: r = −0.635, p = 0.005; alternated 800 ms: r = −0.574, p = 0.013) (see Fig. 8). P-values for multiple comparisons had been adjusted using FDR correction. However, similar correlations were not statistically significant in other conditions.

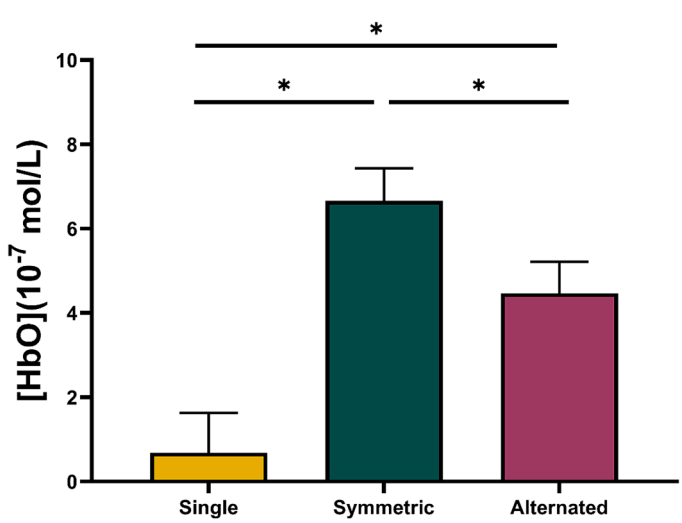

Fig. 7. As for relative phase, [HbO] at the alternated condition was lower than at the symmetric condition, and [HbO] at the single condition was lower than that at the symmetric and alternated condition.

4. Discussion

This study examined the individual deviation in behavioral level and IBS occurred at specific brain regions involved in SMS between individuals under different relative phases and tempos. In terms of individual deviations, performance was better for 400 ms and 800 ms than for 1600 ms, and better for alternated condition than for symmetric condition. Left dlPFC was considered as a key brain region for interpersonal SMS. To explore the mechanisms in depth, we also focused on the interference from partner to indicate the extent to which individuals are influenced by others in their behavior. The symmetric condition exhibited more interference from partners than the alternated condition, whereas the alternated condition showed a higher level of IBS. Furthermore, we explored the relationship between interference in a dyadic task and IBS. Correlation analyses revealed a moderate negative correlation between the interference from partner and IBS under most levels of alternated conditions. This suggests that binary cooperative interaction among individuals towards shared goals may counteract interference from the presence of others, which may contribute to interval subdivision between individuals.

The primary discovery of this study highlights the benefit of maintaining tempo through alternated patterns during SMS in social situations. Firstly, behavioral performance in the two dyadic conditions showed a greater decrease compared to the single condition, suggesting the presence of other people interfering with one's timing, which is in line with previous research findings (Okano, 2017; Kimura, 2020; Kurihara, 2022). Further, although we could not directly show the existence of joint rushing because the individual deviation of the alternated condition was smaller than the symmetric condition but not smaller than the single condition, which makes it puzzling that there was neither joint rushing nor interval subdivision. But we first found at the neural activation level that both dyadic conditions were higher than the single condition in the dlPFC related to cognitive effort, suggesting that the intrinsic mechanisms behind the undifferentiated behavioral outcomes are different for the single and alternated conditions. The alternated condition may have presented higher activation because of the current task tempo, which is difficult to maintain. And the reason why they did not show the same tendency to differ in behavior may be related to the interval subdivision effect. Because of the reference provided to each other and the presence of a common goal, individuals presented a higher IBS between them. Such results support the counterbalancing of the interval subdivision effect for joint rushing rather than an independent SMS process similar to the single condition.

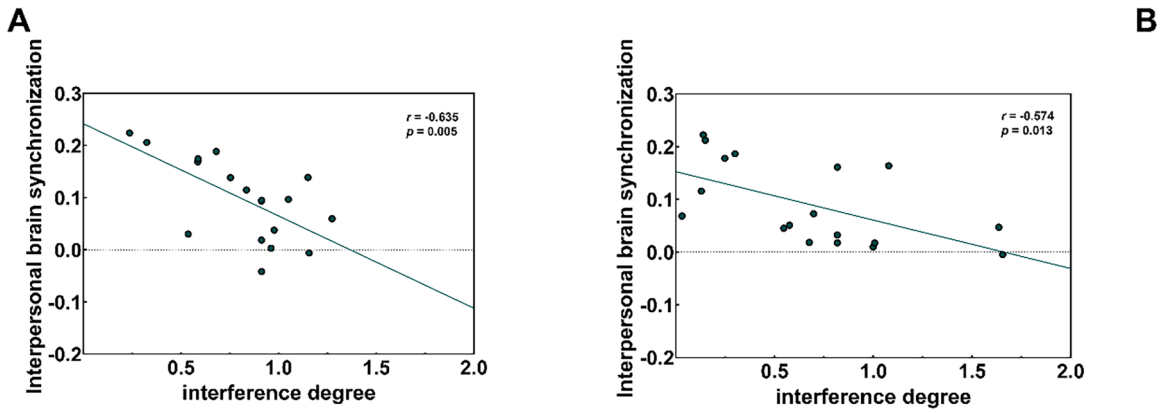

Fig. 8. Correlations between IBS and interference from partner at (A) alternated 400 ms condition and (B) alternated 800 ms condition.

As the pattern of interval subdivision has previously been reflected in the single task, and our results found that individuals segmented intervals of each other and maintained tempos in order to avoid a greater decline through this effect. Although not a double or triple rate tempo presentation of interval subdivision (Patel et al., 2005), the anti-phase form of cutting achieves the same reduction in individual deviation. Furthermore, it can be postulated that individuals in SMS refer to their own tapings and external events (others' tapings) for the next tapping onset adjustment. In the symmetric condition, the decrease in tempo caused the individual to use their own tapping as the primary reference and the external temporal event as the secondary reference (Repp, 2010). This was determined by the relative phases of complete overlap. In contrast, in the alternated condition, external time events can be observed due to the objective in-cycle non-overlapping characteristics of the external time events and the individual's tapping. Consequently, the reference role of external events in the SMS process is maintained to some extent, rather than reduced, in the symmetric condition. This provides objective conditions and possibilities for the creation of interval subdivision. Note that in a dyadic task, individuals are almost impossible to perfectly divide the tapping intervals between each other although we tried to control for this with guidelines and data cleaning. Nevertheless, the IBS and behavioral performance under the alternated condition are better than those under the symmetric condition, indicating that it is not necessary to perfectly divide intervals to induce subdivision effects. This extends previous research that guided individuals to produce subdivision effects using a standard metronome (Ono et al., 2022; Repp, 2003, 2008, 2010; Repp and Jendoubi, 2009). In addition, in our analysis of individual deviation, we have also ruled out a potential possibility, which is that the integer ratio between alternated and symmetric patterns (2/1) implies that moving in alternated fills in the gaps between intervals. This may lead to a collective alternated condition at IOI having the same intervals as the symmetric condition at half of IOI. We found that the differences between symmetric-400ms and alternated-800ms, as well as symmetric-800ms and alternated-1600ms, cannot be observed. This suggests that the more stable timing in alternated patterns cannot be attributed to the reasons mentioned above.

In addition, SMS at different IOIs showed different timing patterns. Individuals' tempo maintenance performance was better, and no difference existed between them at the 400 and 800 ms IOIs. However, there was a significant drop in tempo maintenance at 1600 ms IOI. This drop may be due to the increased cognitive effort required to maintain tapping performance after 1500 ms IOI (Miyake et al., 2004). The behavioral difference between 400 and 800 ms was slight and, therefore, insufficient to capture differences in behavioral performance. We found features consistent with behavioral performance in IBS, which IBS was higher than 1600 ms on both 400 and 800 ms, whereas there was no difference between 400 ms and 800 ms. Meanwhile, a negative correlation between IBS and the interference from partner can only be observed in the alternated condition at 400 and 800 ms. The combination of these results suggests that alternated condition with phase advantage can help individuals to some extent counteract interference from partners. Furthermore, we did not find any differences in the interference from partner between different tempos, which, in combination with the decline in cliff-edge performance below 1600 ms, suggests that the length of the interval may only be a challenge to individuals' internal timing, rather than influencing mutual interference between individuals. The benefits of the subdivision effect generated by the alternated condition may mainly manifest in interference among individuals rather than internal interference within individuals.

Regarding the involvement of functional brain regions, we discovered an increased IBS in the left dlPFC. To our knowledge, few previous studies have identified and highlighted the role of dlPFC in interval subdivision effects between two brains which involved in interpersonal SMS. During collaborative problem - solving activities or joint drawing, the dlPFC displays enhanced IBS (Li et al., 2021; Lu and Hao, 2019; Lu et al., 2019a; Lu et al., 2019b; Sun et al., 2020; Xue et al., 2018). One of the main functions of the dlPFC is related to cognitive flexibility associated with attentional switching (Monsell, 2003). In tasks involving individuals' perception of synchronized movement, attentional shifting between partner and self-assigned tasks is required. The IBS in the dlPFC may serve as the foundation for this flexible attentional switching. This is also consistent with the mutual prediction theory, in which interpersonal interactions may facilitate the reconfiguration of intra- and inter-brain activities. The reorganization of these intra- and inter-brain networks constitutes the basis for behavioral coordination during interactive processes. The increased involvement of the IBS, including the dlPFC, reflects an enhancement of this mutual predictive trend (Mayo and Shamay-Tsoory, 2024). Our study further provides promising assessment methods for individuals undergoing rehabilitation for movement disorders or stroke. It raises the question of whether individuals can generate and use movement information to guide their timing. We seek to establish how extraneous information may impact gait, leading to disturbances, and how to counteract it. This determines if an individual's capacity to perceive time about working memory is impaired based on greater ecological validity.

We simultaneously observed the individual activation results of the dlPFC, where compared to the alternated condition, more dlPFC recruitment occurred in the symmetric condition. Meanwhile, activation in two dyadic conditions were both higher than single condition. In terms of activation levels, dlPFC is associated with brain regions related to attention resource allocation and cognitive flexibility (Tei et al., 2017; White et al., 2023; Wu et al., 2014). Watanabe et al. (2018) found a positive correlation between dlPFC activation and self-reported cognitive effort. Additionally, dlPFC is related to working memory, which is required in time perception (Lewis and Miall, 2006; Ustun et al., 2017;

Vicario, 2013). Although direct measurement of cognitive effort was not conducted, our findings regarding the activation of dlPFC may suggest that: 1) compared to single SMS, the dyadic task produced higher activation in response to new demands for cognitive resources outside of SMS within individuals. Combined with the higher individual deviation in the dyadic conditions, this may be a bigger challenge posed by joint rushing; 2) the distinction of IBS between relative phase in the dyadic condition, coupled with similar patterns in individual deviation and interference from others, leads to the presence of an SMS advantage in the alternated condition. This may provide further evidence for the advantage of interval subdivision in the dyadic condition. Future research could further investigate how cognitive resources are involved in interpersonal SMS through self-report or dyadic task paradigms.

One limitation of our study is that tempo is treated as a discrete variable, making it difficult to observe changes in the proportions of internal timings and external interference that correspond to the timing bias presented by individuals as the interval increases. Rosso et al. (2021b and 2023b) using the drift metronome is an effective method to explore all relative phases between partners, maximizing the understanding of the mechanisms of dyadic coordination between individuals under different relative phases. This may be a good way to address the current research gaps. Future research may examine the activation traits of brain regions connected to attentional allocation by constantly modifying relative phases or IOIs. Furthermore, for the occurrence of joint rushing, the model of Wolf et al. (2019) proposes a phase advancement mechanism. This mechanism suggests that joint rushing is more pronounced when different individuals produce perceptually similar action effects. Alternatively, individuals have a sensitive window and will only shorten their cycle if a signal from others enters their sensitive window. The current study cannot directly determine whether the alternative condition produced exactly the same joint rushing with symmetric condition although we have speculated by combining neural data at both the activation and interbrain levels. Future studies could attempt to calculate the exact extent of the sensitive window for joint rushing.

In summary, our study combines behavioral and neural indicators to discover the application of the interval subdivision effect in social situations and highlights the significant role of dlPFC. In addition, we support and imply differences in the mechanisms underlying timing at different tempos. This expands the application of relative phase differences in traditional dynamics studies.

Consent to participate

Written informed consent was obtained from all individual participants included in the study. The privacy rights of human subjects were always be observed.

Data and code availability statements

Data available on request from the authors.

CRediT authorship contribution statement

Ruoyu Niu: Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Xiaodan Xu:** Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Weicai Tang:** Data curation. **Yi Xiao:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Rixin Tang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

All the authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

Ethical approval

The study followed the ethical guidelines of the Declaration of Helsinki and was approved by the Ethical Committee of the Department of Psychology, Nanjing University (No. NJUPSY202205004).

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2024.120788.

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